NETWORK SECURITY PRACTICES – ATTACK AND DEFENSE

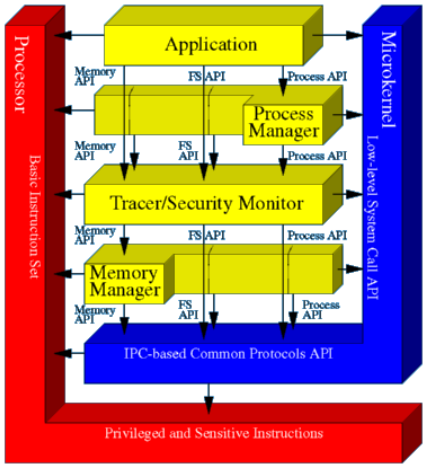
SELinux

Overview

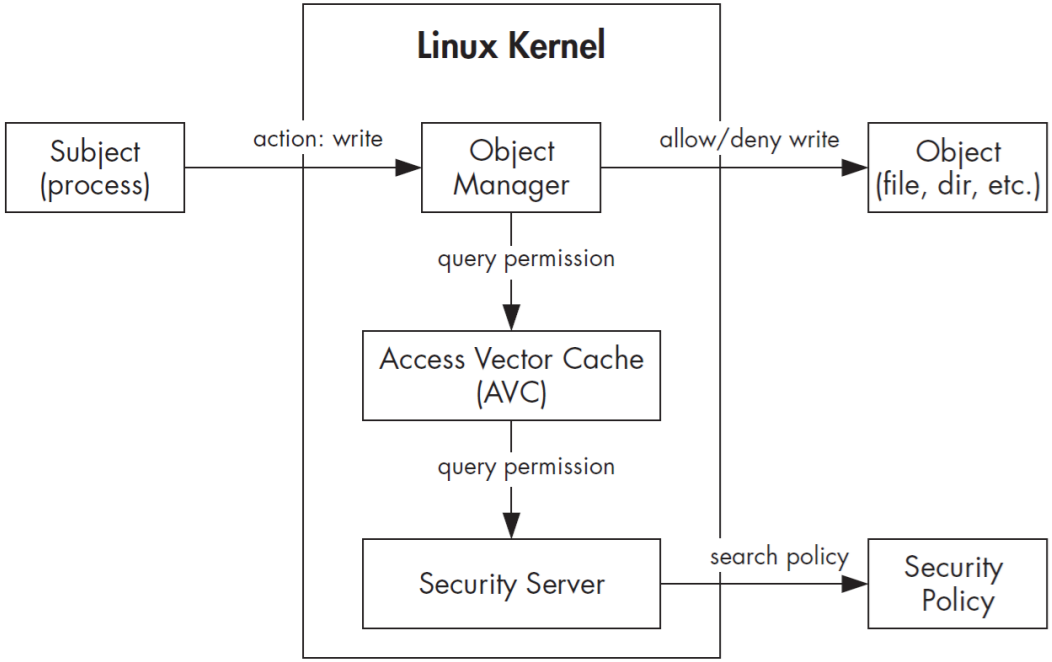
- Originally from NSA
- Mandatory Access Control for Linux
  - Security contexts
    - Users
    - Roles
    - Types (domains)
    - Security level (MLS)
  - Policies
- Unprivileged root user
  - If a process which is running as the root user on an SELinux system is compromised (such as through a buffer overflow, etc.) then the damage is limited to what that process can access as defined by the policy

Overview

- Distributed Trusted Mach by NSA in 1992
  - Type enforcement in the Mach microkernel
- Distributed Trusted Operating System (DTOS) by NSA
- University of Utah's Fluke OS
  - http://www.cs.utah.edu/flux/fluke/html/index.html
- DTOS + Fluke => Flask
  - http://www.cs.utah.edu/flux/fluke/html/flask.html
  - Flask then become SELinux



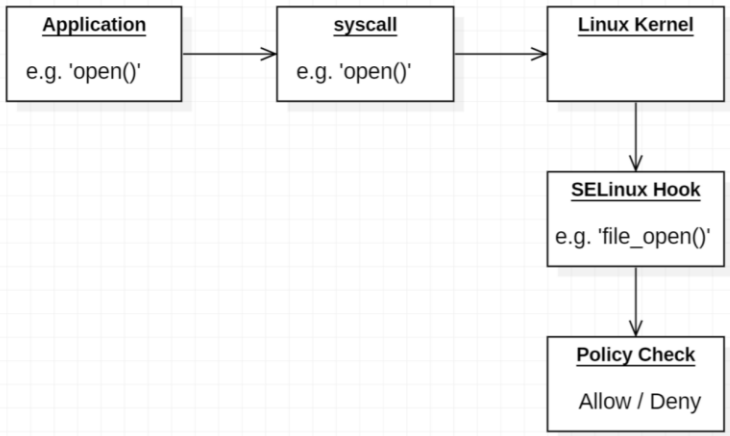
Overview



- SELinux Tools
  - yum install selinux-policy
  - yum install selinux-policy-devel
  - yum install policycoreutils
  - yum install setools-console

Overview

```
// security/selinux/hooks.c
static int selinux_file_open(struct file *file)
{
    // 1. Get current process's security ID (SID)
    u32 sid = current_sid();

    // 2. Get file's security context from inode
    struct inode *inode = file_inode(file);
    u32 fsid = inode->i_security;

    // 3. Check permissions through Access Vector Cache
    int rc = avc_has_perm(sid,                // Subject SID
                          fsid,               // Object SID
                          SECCLASS_FILE,      // Object class
                          FILE__OPEN,         // Requested
permission
                          NULL);              // Audit data

    // 4. Return decision (0=ALLOW, negative=DENY)
    return rc;
}
```



Simplified selinux_file_open in Android kernel source (courtesy by Stefano Santilli)

File system labels

```
[hank@localhost /]$ ls -lZ
lrwxrwxrwx. root root system_u:object_r:bin_t:s0       bin -> usr/bin
dr-xr-xr-x. root root system_u:object_r:boot_t:s0      boot
drwxr-xr-x. root root system_u:object_r:device_t:s0    dev
drwxr-xr-x. root root system_u:object_r:etc_t:s0       etc
drwxr-xr-x. root root system_u:object_r:home_root_t:s0 home
lrwxrwxrwx. root root system_u:object_r:lib_t:s0       lib -> usr/lib
lrwxrwxrwx. root root system_u:object_r:lib_t:s0       lib64 -> usr/lib64
drwx------. root root system_u:object_r:lost_found_t:s0 lost+found
drwxr-xr-x. root root system_u:object_r:mnt_t:s0       media
drwxr-xr-x. root root system_u:object_r:mnt_t:s0       mnt
drwxr-xr-x. root root system_u:object_r:usr_t:s0       opt
dr-xr-xr-x. root root system_u:object_r:proc_t:s0      proc
dr-xr-x---. root root system_u:object_r:admin_home_t:s0 root
drwxr-xr-x. root root system_u:object_r:var_run_t:s0   run
lrwxrwxrwx. root root system_u:object_r:bin_t:s0       sbin -> usr/sbin
drwxr-xr-x. root root system_u:object_r:var_t:s0       srv
dr-xr-xr-x. root root system_u:object_r:sysfs_t:s0     sys
drwxrwxrwt. root root system_u:object_r:tmp_t:s0       tmp
drwxr-xr-x. root root system_u:object_r:usr_t:s0       usr
drwxr-xr-x. root root system_u:object_r:var_t:s0       var
[hank@localhost /]$
```

user role type level

Process labels

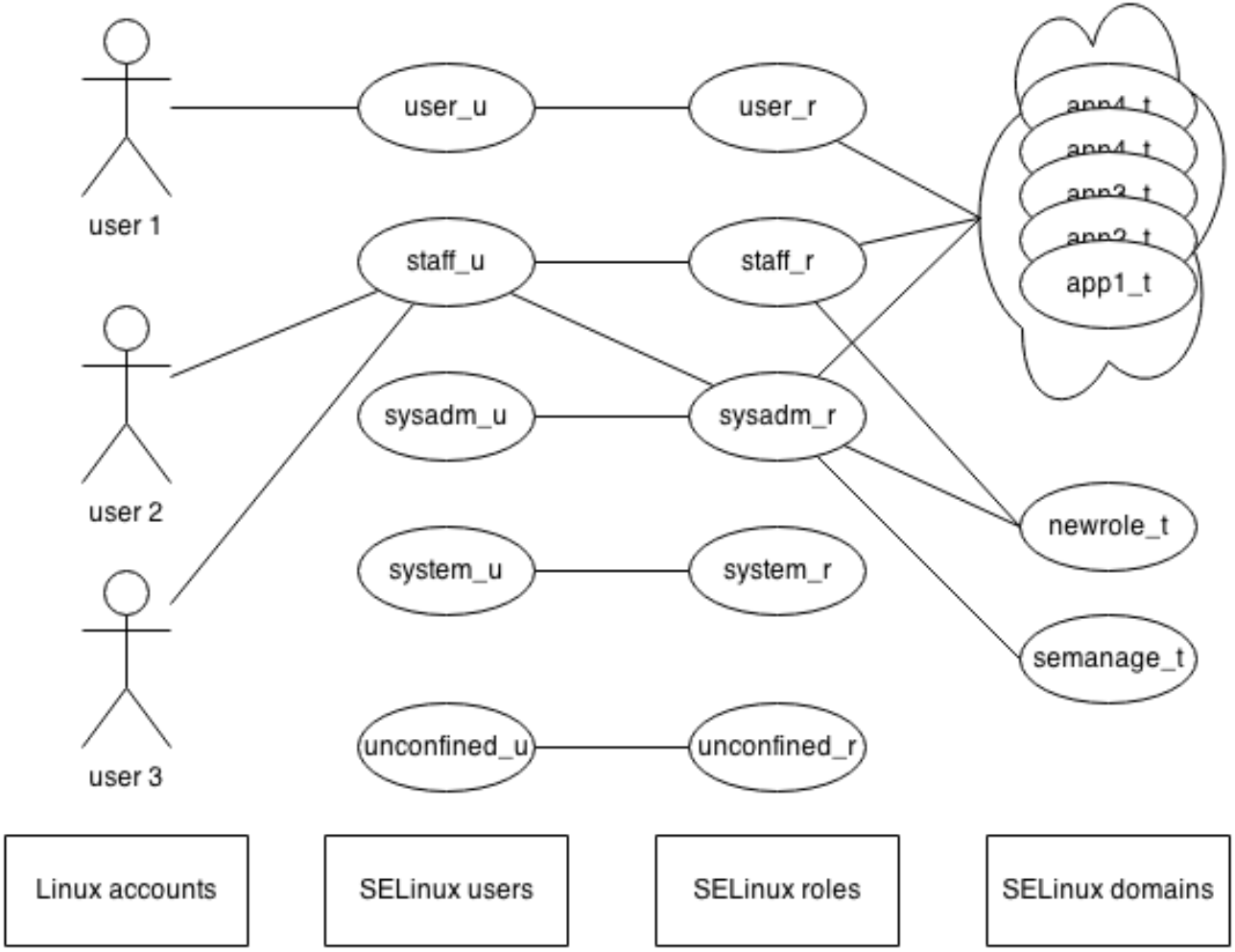
```
[hank@localhost ~]$ ps Z
LABEL                              PID TTY      STAT   TIME COMMAND
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 4622 pts/0 Ss   0:00 bash
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 4657 pts/0 R+   0:00 ps Z
[hank@localhost ~]$
```

user

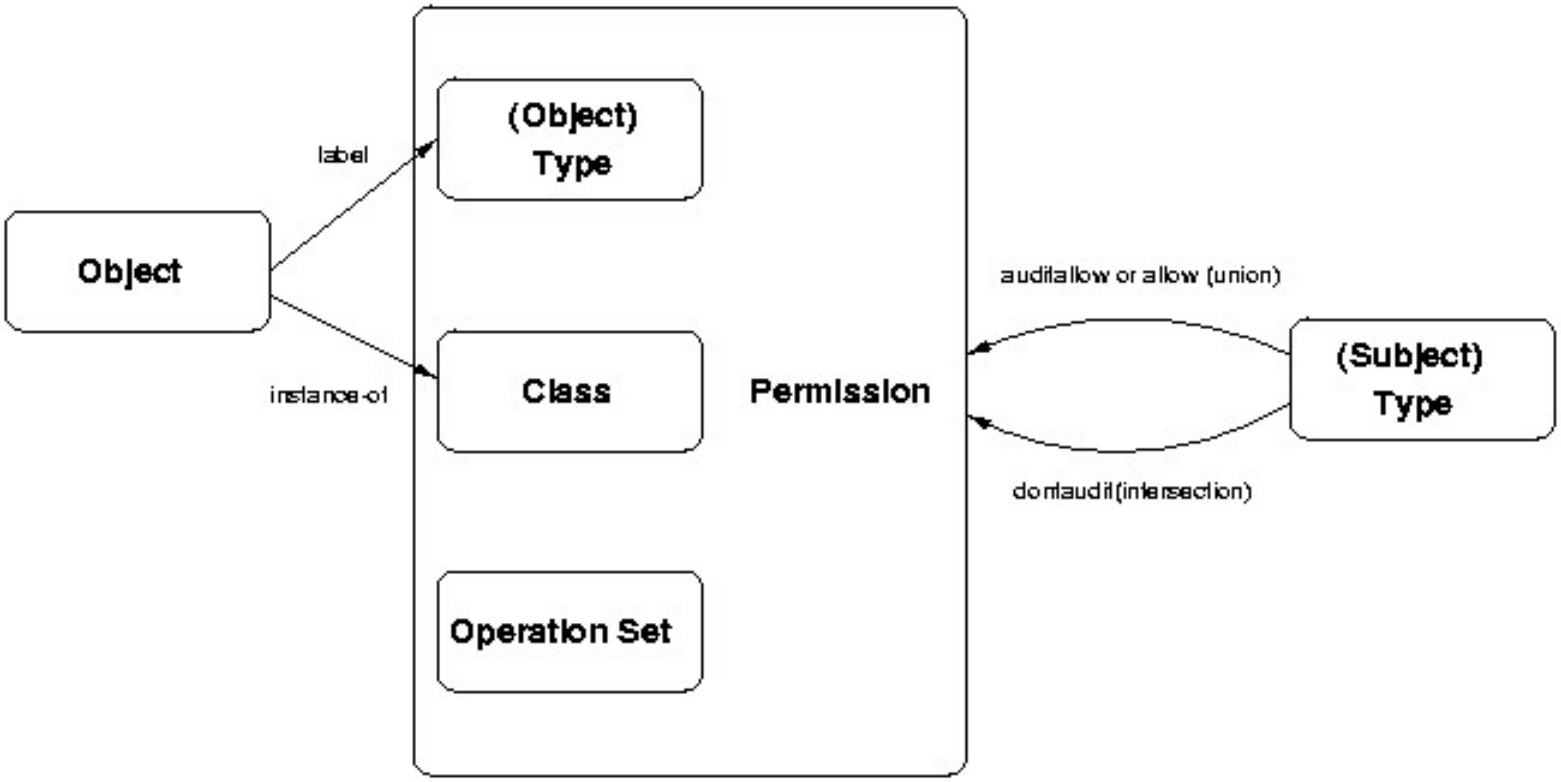
role

domain (type)

level


user 1
user 2
user 3
user_u
staff_u
sysadm_u
system_u
unconfined_u
user_r
staff_r
sysadm_r
system_r
unconfined_r
app1_t
newrole_t
semanage_t
Linux accounts
SELinux users
SELinux roles
SELinux domains

SELinux Type Enforcement



Type enforcement

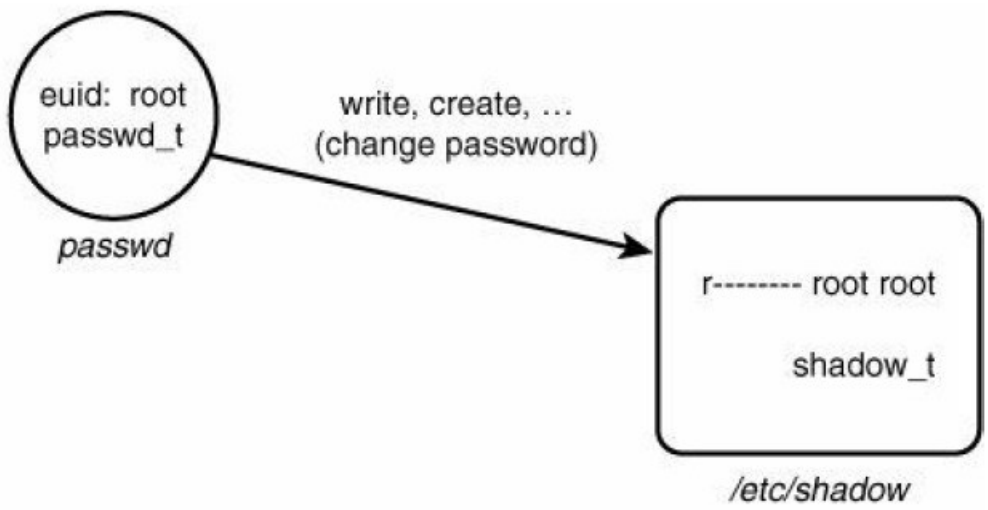
- rule_name source_type target_type : class perm_set;

```
[root@better homeworks]# sesearch -s httpd_t -ACTS | more
Found 1311 semantic av rules:
   allow httpd_t security_t : filesystem getattr ;
   allow httpd_t security_t : file { ioctl read getattr lock open } ;
   allow httpd_t security_t : dir { ioctl read getattr lock search open } ;
   allow httpd_t security_t : lnk_file { read getattr } ;
   allow nsswitch_domain nmbd_var_run_t : dir { getattr search open } ;
   allow nsswitch_domain nmbd_var_run_t : sock_file { write getattr append open } ;
   allow nsswitch_domain nslcd_var_run_t : dir { getattr search open } ;
   allow nsswitch_domain nslcd_var_run_t : sock_file { write getattr append open } ;
   allow corenet_unlabeled_type unlabeled_t : tcp_socket recvfrom ;
   allow corenet_unlabeled_type unlabeled_t : udp_socket recvfrom ;
   allow corenet_unlabeled_type unlabeled_t : rawip_socket recvfrom ;
   allow syslog_client_type device_t : dir { ioctl read getattr lock search open } ;
   allow syslog_client_type device_t : lnk_file { read getattr } ;
   allow syslog_client_type devlog_t : lnk_file { read getattr } ;
   allow syslog_client_type devlog_t : sock_file { write getattr append open } ;
   allow corenet_unlabeled_type unlabeled_t : association { sendto recvfrom } ;
   allow corenet_unlabeled_type unlabeled_t : dccp_socket recvfrom ;
   allow corenet_unlabeled_type unlabeled_t : peer recv ;
```

.....

Type enforcement

Figure 2-2. Type enforcement example: passwd program



allow passwd_t shadow_t : file {ioctl read write create getattr setattr lock relabelfrom relabelto append unlink link rename};

(from SELinux by Example - Using Security Enhanced Linux)

Create your own policy

- Write a TE policy

~/selinux_test/policy/test.te – Sublime Text 2 (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

test.te

```
policy_module(test, 1.0)

gen_require(`
    type user_t;
    type var_log_t;
')

allow user_t var_log_t:dir {getattr search open read};
```

Line 8, Column 55 Tab Size: 4 Plain Text

Create your own policy

- Install policy makefile
  - yum install selinux-policy-devel
- Compile policy

```
[hank@localhost policy]$ make -f  /usr/share/selinux/devel/Makefile test.pp
Compiling targeted test module
/usr/bin/checkmodule:  loading policy configuration from tmp/test.tmp
/usr/bin/checkmodule:  policy configuration loaded
/usr/bin/checkmodule:  writing binary representation (version 17) to tmp/test.mod
Creating targeted test.pp policy package
rm tmp/test.mod.fc tmp/test.mod
[hank@localhost policy]$
```

- Install policy
  - semodule –i test.pp

File type and process type

hank@localhost:~/selinux_test

File Edit View Search Terminal Help

```
[hank@localhost selinux_test]$ cat a.c
#include <stdio.h>

int main()
{

        printf("Hello World...press enter to continue\n");
        getchar();
        return 0;
}
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$ ls -lZ
-rw-rw-r--. hank hank unconfined_u:object_r:user_home_t:s0 a.c
-rwxrwxr-x. hank hank unconfined_u:object_r:user_home_t:s0 a.out
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$ ./a.out
Hello World...press enter to continue
```

executable file

Type transition

hank@localhost:~/selinux_test

File Edit View Search Terminal Help

process

```
[hank@localhost selinux_test]$ ps aZ|grep a.out
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 3766 pts/0 S+   0:00 ./a.out
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 3839 pts/1 S+   0:00 grep --color=auto a.out
[hank@localhost selinux_test]$

[hank@localhost selinux_test]$
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$
```

File type and process type

~/selinux_test/policy/test.te – Sublime Text 2 (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

test.te

```
1  policy_module(test, 1.0)
2
3  gen_require(`
4      type user_t;
5      type var_log_t;
6  ')
7
8  type test_t;
9  type test_exec_t;
10
11 allow user_t var_log_t:dir {getattr search open read};
```

Define a new process type

Define a new file type

2 lines, 30 characters selected Tab Size: 4 Plain Text

Relabeling file type

```
[hank@localhost selinux_test]$ ls -al
total 20
drwxrwxr-x.  3 hank hank   41 Dec 14 08:57 .
drwx------. 15 hank hank 4096 Dec 14 08:59 ..
-rw-rw-r--.  1 hank hank  111 Dec 14 08:44 a.c
-rwxrwxr-x.  1 hank hank 8560 Dec 14 08:44 a.out
drwxrwxr-x.  3 hank hank   72 Dec 14 09:11 policy
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$ chcon -t test_exec_t a.out
chcon: failed to change context of 'a.out' to 'unconfined_u:object_r:test_exec_t:s0': Permission denied
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$


[hank@localhost selinux_test]$


[hank@localhost selinux_test]$
```

Relabeling file type

- Check the audit log (/var/log/audit/audit.log) for access violations
- unconfined_t does not have relabelto permission to test_exec_t:file

```
[root@localhost audit]# tail audit.log
type=LOGIN msg=audit(1418567401.198:454): pid=5106 uid=0 subj=system_u:system_r:crond_t:s0-s0:c0.c1023 old-auid=4294967295 auid=0 old-
ses=4294967295 ses=7 res=1
type=USER_AVC msg=audit(1418567401.214:455): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='avc:  re
ceived policyload notice (seqno=2)  exe="/usr/lib/systemd/systemd" sauid=0 hostname=? addr=? terminal=?'
type=USER_START msg=audit(1418567401.220:456): pid=5106 uid=0 auid=0 ses=7 subj=system_u:system_r:crond_t:s0-s0:c0.c1023 msg='op=PAM:s
ession_open acct="root" exe="/usr/sbin/crond" hostname=? addr=? terminal=cron res=success'
type=CRED_REFR msg=audit(1418567401.222:457): pid=5106 uid=0 auid=0 ses=7 subj=system_u:system_r:crond_t:s0-s0:c0.c1023 msg='op=PAM:se
tcred acct="root" exe="/usr/sbin/crond" hostname=? addr=? terminal=cron res=success'
type=CRED_DISP msg=audit(1418567401.234:458): pid=5106 uid=0 auid=0 ses=7 subj=system_u:system_r:crond_t:s0-s0:c0.c1023 msg='op=PAM:se
tcred acct="root" exe="/usr/sbin/crond" hostname=? addr=? terminal=cron res=success'
type=USER_END msg=audit(1418567401.237:459): pid=5106 uid=0 auid=0 ses=7 subj=system_u:system_r:crond_t:s0-s0:c0.c1023 msg='op=PAM:ses
sion_close acct="root" exe="/usr/sbin/crond" hostname=? addr=? terminal=cron res=success'
type=MAC_POLICY_LOAD msg=audit(1418567563.351:460): policy loaded auid=1000 ses=1
type=SYSCALL msg=audit(1418567563.351:460): arch=c000003e syscall=1 success=yes exit=3591088 a0=4 a1=7f590c911010 a2=36cbb0 a3=7fff17b
6f860 items=0 ppid=5196 pid=5197 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=1 comm="load_policy" e
xe="/usr/sbin/load_policy" subj=unconfined_u:system_r:unconfined_t:s0-s0:c0.c1023 key=(null)
type=AVC msg=audit(1418567674.029:461): avc:  denied  { relabelto } for  pid=5262 comm="chcon" name="a.out" dev="dm-2" ino=188 scontex
t=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 tcontext=unconfined_u:object_r:test_exec_t:s0 tclass=file
type=SYSCALL msg=audit(1418567674.029:461): arch=c000003e syscall=188 success=no exit=-13 a0=25d10e0 a1=7f707576c79e a2=25d2610 a3=25
items=0 ppid=4100 pid=5262 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000 tty=pts2 ses=1 co
mm="chcon" exe="/usr/bin/chcon" subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(null)
[root@localhost audit]#
```

- Automatic policy generation from audit log
  - audit2allow –i /var/log/audit/audit.log –o /tmp/mypolicy.te

Relabeling file type

```
policy_module(test, 1.0)

gen_require(`
   type user_t;
   type unconfined_t;
   type var_log_t;
')

type test_t;
type test_exec_t;

allow user_t var_log_t:dir {getattr search open read};
allow unconfined_t test_exec_t:file relabelto;
```

Relabeling file type

```
drwxrwxr-x.  3 hank hank   41 Dec 14 08:57 .
drwx------. 15 hank hank 4096 Dec 14 08:59 ..
-rw-rw-r--.  1 hank hank  111 Dec 14 08:44 a.c
-rwxrwxr-x.  1 hank hank 8560 Dec 14 08:44 a.out
drwxrwxr-x.  3 hank hank   72 Dec 14 09:11 policy
[hank@localhost selinux_test]$ chcon -t test_exec_t a.out
chcon: failed to change context of ‘a.out’ to ‘unconfined_u:object_r:test_exec_t:s0’: Permission denied
[hank@localhost selinux_test]$
```

Relabeling file type

```
policy_module(test, 1.0)

gen_require(`
   type user_t;
   type unconfined_t;
   type var_log_t;
   type fs_t;
')

type test_t;
type test_exec_t;

allow user_t var_log_t:dir {getattr search open read};
allow unconfined_t test_exec_t:file relabelto;
allow test_exec_t fs_t:filesystem associate;
```

Type transition – relabeling file type

hank@localhost:/home/hank/selinux_test

File Edit View Search Terminal Help

```
[root@localhost selinux_test]# ls -lZ
ls: cannot access a.out: Permission denied
-rw-rw-r--. hank hank unconfined_u:object_r:user_home_t:s0 a.c
?---------  ?    ?                                         a.out
drwxrwxr-x. hank hank unconfined_u:object_r:user_home_t:s0 policy
[root@localhost selinux_test]# ls -lZ
-rw-rw-r--. hank hank unconfined_u:object_r:user_home_t:s0 a.c
-rwxrwxr-x. hank hank unconfined_u:object_r:test_exec_t:s0 a.out
drwxrwxr-x. hank hank unconfined_u:object_r:user_home_t:s0 policy
[root@localhost selinux_test]# ./a.out
bash: ./a.out: Permission denied
[root@localhost selinux_test]#
[root@localhost selinux_test]#
```

`ls` shows nothing here, why?

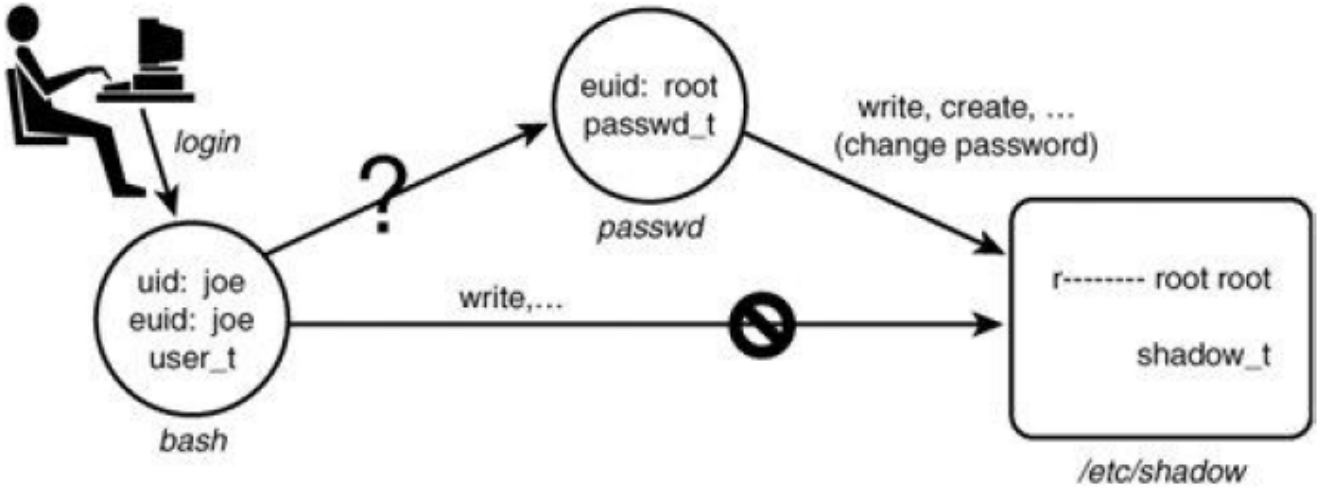
Relabeling file type

```
policy_module(test, 1.0)

gen_require(`
   type user_t;
   type unconfined_t;
   type var_log_t;
   type fs_t;
')

type test_t;
type test_exec_t;

allow user_t var_log_t:dir {getattr search open read};
allow unconfined_t test_exec_t:file relabelto;
allow test_exec_t fs_t:filesystem associate;
allow unconfined_t test_exec_t:file getattr;
```

Type transition for processes

Figure 2-3. The problem of domain transitions

[View full size image]

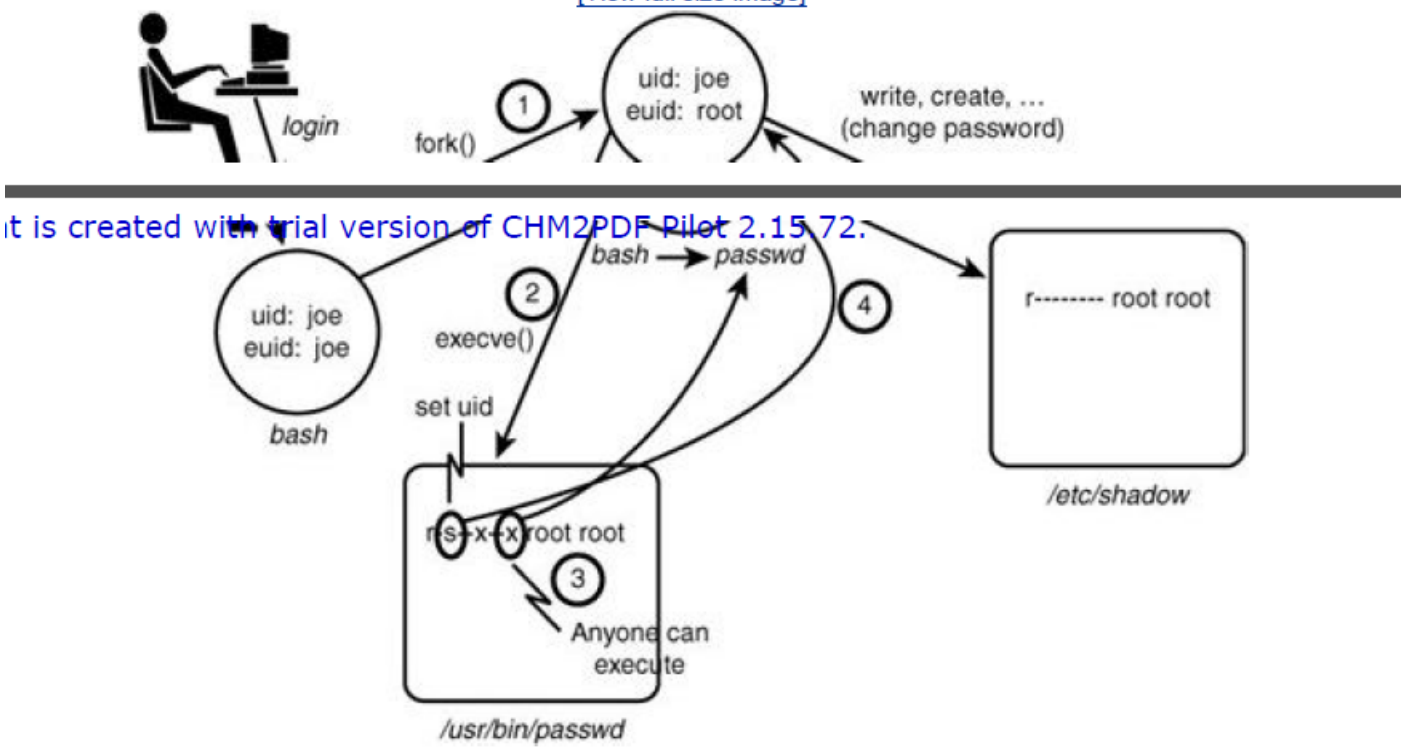


(from SELinux by Example - Using Security Enhanced Linux)

Type transition for processes

Figure 2-4. Password program security in standard Linux (setuid)

[View full size image]

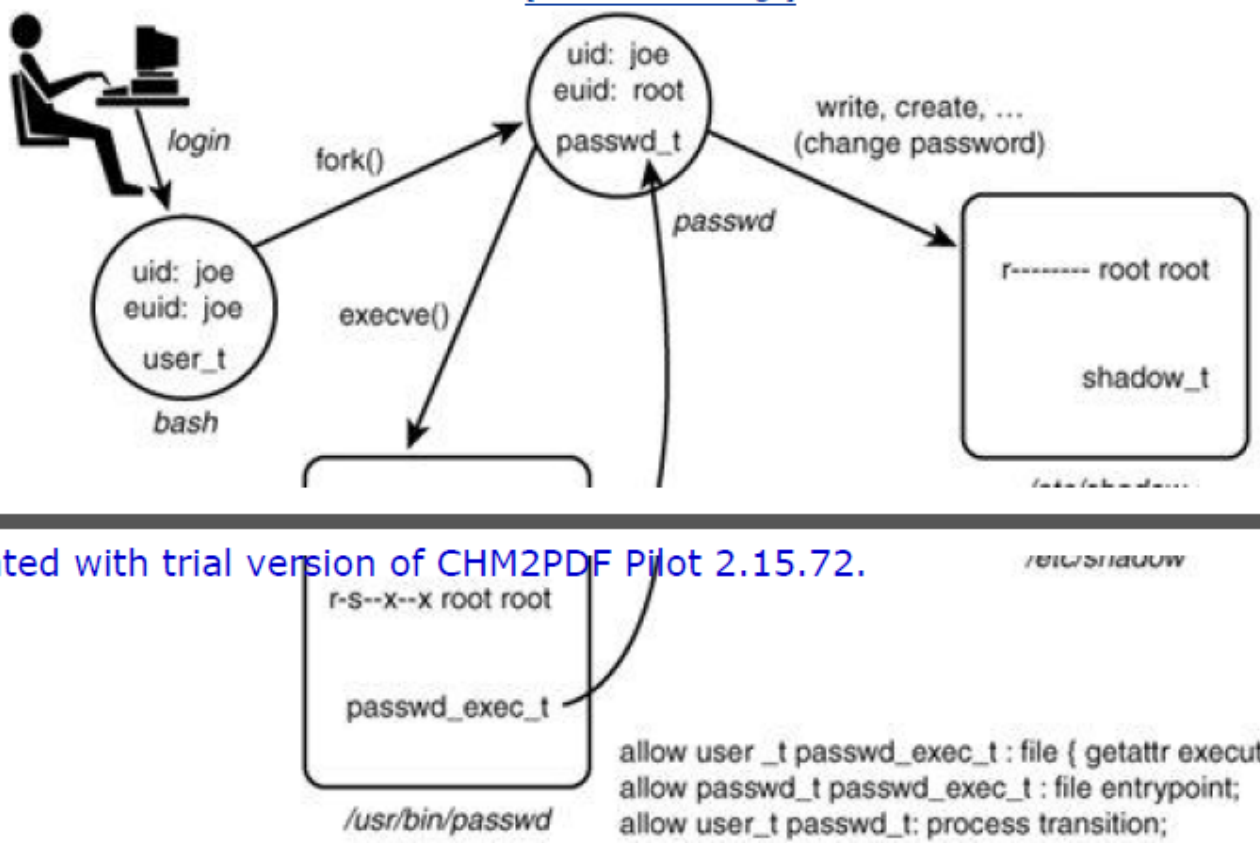


It is created with trial version of CHM2PDF Pilot 2.15.72.

Type transition for processes

Figure 2-5. Passwd program security in SELinux (domain transitions)

[View full size image]



nt is created with trial version of CHM2PDF Pilot 2.15.72.

Type transition for processes

- SELinux by default inherits contexts, be it from processes (on fork) or parent files/directories
- Contexts of processes can change on execute of a command from that process' context, but only under the conditions that
  - The target file context is executable for the source domain
  - The target file context is marked as an entrypoint for the target domain
  - The source domain is allowed to transition to the target domain
- Macro for process type transition
  - domain_auto_trans(source, target, new_type);
  - E.g.
    - `domain_auto_trans(sshd_t, shell_exec_t, user_t);`

Type transition for processes (example)

hank@localhost:~/selinux_test

File Edit View Search Terminal Help

```
[hank@localhost selinux_test]$ ls -lZ
-rw-rw-r--. hank hank unconfined_u:object_r:user_home_t:s0 a.c
-rwxrwxr-x. hank hank unconfined_u:object_r:test_exec_t:s0 a.out
drwxrwxr-x. hank hank unconfined_u:object_r:user_home_t:s0 policy
[hank@localhost selinux_test]$
[hank@localhost selinux_test]$ ./a.out
Hello World...press enter to continue
```

hank@localhost:/home/hank/selinux_test/policy

File Edit View Search Terminal Help

```
[root@localhost policy]# ps aZ|grep a.out
unconfined_u:unconfined_r:test_t:s0-s0:c0.c1023 13994 pts/1 S+   0:00 ./a.out
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 13997 pts/0 S+   0:00 grep --color=auto a.out
[root@localhost policy]#
```

Type transition for processes (example)

```
policy_module(test, 1.0)

gen_require(`
   type unconfined_t;
   type fs_t;
   type null_device_t;
   type user_devpts_t;
   type root_t;
   type etc_t;
   type lib_t;
   type ld_so_cache_t;
   type ld_so_t;
   type usr_t;
   type kernel_t;
   type ksmtuned_t;
   role unconfined_r;
')

type test_t;
type test_exec_t;

type_transition unconfined_t  test_exec_t:
                                 process  test_t;


#============= ksmtuned_t ==============
allow ksmtuned_t test_t:dir { getattr search };
allow ksmtuned_t test_t:file { read open };
```

```
#============= unconfined_t ==============
allow unconfined_t test_exec_t:file {relabelto getattr
                                      execute  read open};
allow unconfined_t test_t:process { siginh rlimitinh noatsecure
                                    getattr};
allow unconfined_t test_t :  process transition;
allow unconfined_t test_t:dir { getattr search };
allow unconfined_t test_t:file { read open };
allow unconfined_t test_t:lnk_file read;

#============= test_exec_t ==============
allow test_exec_t fs_t:filesystem associate;


#============= test_t ==============
allow test_t test_exec_t: file entrypoint;
allow test_t unconfined_t:process sigchld;
allow test_t null_device_t:chr_file { read write open };
allow test_t user_devpts_t:chr_file { read write getattr};
allow test_t unconfined_t:fd use;
allow test_t root_t:dir search;
allow test_t etc_t:dir search;
allow test_t lib_t:lnk_file read;
allow test_t lib_t:dir { search getattr };
allow test_t lib_t:file {open read getattr execute};
allow test_t ld_so_cache_t:file {open read getattr};
allow test_t usr_t:dir search;
allow test_t test_exec_t:file read;
allow test_t ld_so_t:file {open read getattr};

#============= kernel_t ==============
allow kernel_t test_t:dir search;

role unconfined_r types test_t;
```

Type transition for objects

- Every created object has a default context
- For example, files and directories are created by default with their parent directory's context
- It's often desirable that the type of the new object will depend on who created it. ``Who'' means what domain (type) the process had.
- For example: If the X server creates a file in the `/tmp` directory, it should have type `xdm_tmp_t`, but if a ``normal user'' process does so, it should be `user_tmp_t`

(from: http://billauer.co.il/selinux-policy-module-howto.html)

Type transitions for objects

```
type_transition Source Target:Class new_type;
```

- This means ``any object of class Class, which is created by a process in the domain (type) Source, and would by default get the type Target, will get the type new_type instead''

Example:

```
type_transition sshd_t tmp_t:file sshd_tmp_t;
```

- ... which means that if a process running in the `sshd_t` domain (most likely the ssh deamon) creates a plain regular file which should have gotten the `tmp_t` type (most likely because it's in the `/tmp` directory), it should get the `sshd_tmp_t` instead.

- Note that this is not a permission rule. Rather, this tells SELinux itself to perform an action.
- The type transition for objects doesn't require an additional permission rule
- But several other actions need permission:
- Read-write access to the parent directory
- Creating a new file or directory with the new type
- To make things easier, a macro bundles the type transition statement with the permissions, `file_type_auto_trans`
- Paraphrasing the last example, the following macro statement covers a variety of file types (plain files, directories, symlinks etc) and also handles the permissions. All in one:

(from: http://billauer.co.il/selinux-policy-module-howto.html)

Unprivileged user

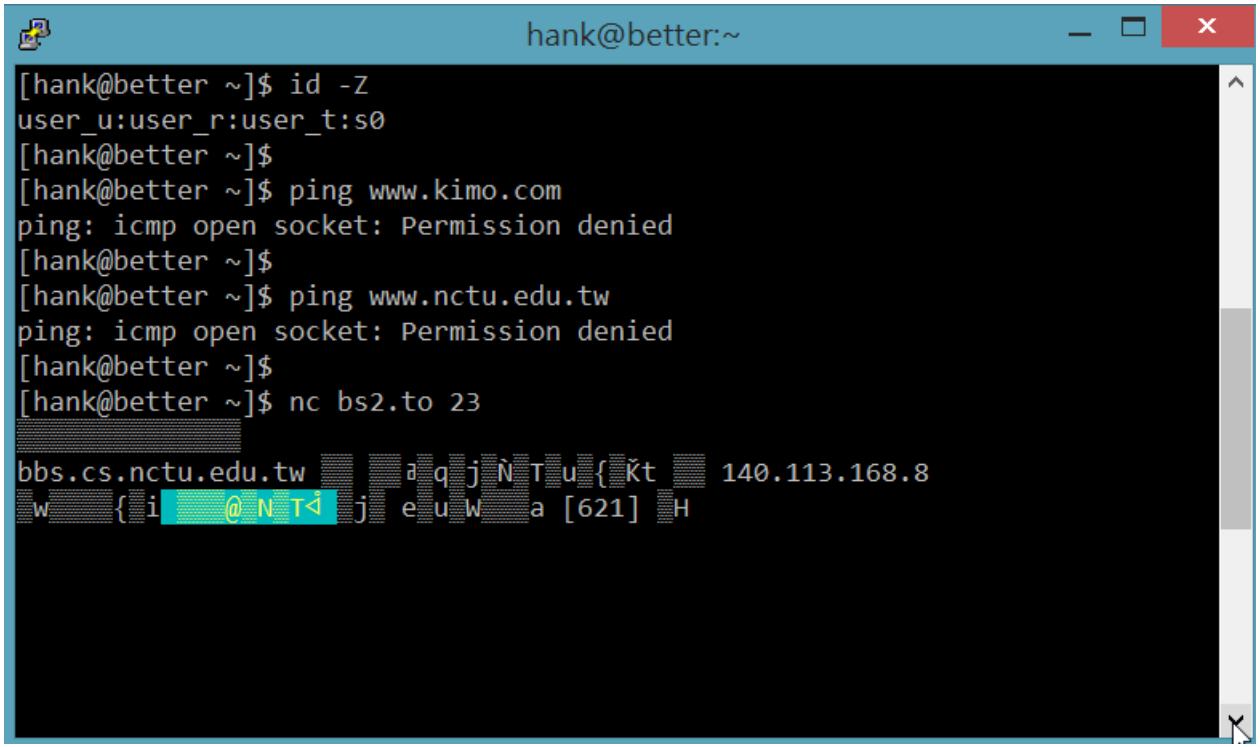
- Disable *ping* for guest user *hank*

hank@better:~

```
[hank@better ~]$ id -Z
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[hank@better ~]$
[hank@better ~]$ ping www.nctu.edu.tw
PING www.nctu.edu.tw (140.114.60.211) 56(84) bytes of data.
64 bytes from www-nctu.et.nthu.edu.tw (140.114.60.211): icmp_seq=1 ttl=58 time=1.66 ms
64 bytes from www-nctu.et.nthu.edu.tw (140.114.60.211): icmp_seq=2 ttl=58 time=1.53 ms
64 bytes from www-nctu.et.nthu.edu.tw (140.114.60.211): icmp_seq=3 ttl=58 time=24.0 ms
64 bytes from www-nctu.et.nthu.edu.tw (140.114.60.211): icmp_seq=4 ttl=58 time=16.5 ms
```

Unprivileged user

```
[root@better homeworks]# semanage login -a -s user_u hank
[root@better homeworks]# setsebool user_ping off
[root@better homeworks]#
```



Unprivileged user

```
[root@better homeworks]# setsebool user_ping on
```

hank@better:~

```
[hank@better ~]$ ping www.kimo.com
PING tw-src.g03.yahoodns.net (124.108.105.150) 56(84) bytes of data.
64 bytes from w2.src.vip.tw1.yahoo.com (124.108.105.150): icmp_seq=1 ttl=53
 time=3.40 ms
64 bytes from w2.src.vip.tw1.yahoo.com (124.108.105.150): icmp_seq=2 ttl=53
 time=3.44 ms
64 bytes from w2.src.vip.tw1.yahoo.com (124.108.105.150): icmp_seq=3 ttl=53
 time=3.55 ms
```

root@better:/

```
[root@better /]# ps auxZ|grep hank
system_u:system_r:sshd_t:s0-s0:c0.c1023 root 15111 0.0  0.0 135520 5240 ?      Ss   Dec08   0:00 sshd: hank [priv]
user_u:user_r:user_t:s0         hank     15119  0.0  0.0 135520  2212 ?        S    Dec08   0:00 sshd: hank@pts/1
user_u:user_r:user_t:s0         hank     15127  0.0  0.0 116412  3188 pts/1    Ss   Dec08   0:00 -bash
user_u:user_r:ping_t:s0         hank     15272  0.0  0.0 120700  1116 pts/1    S+   Dec08   0:00 ping www.kimo.com
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 root 15408 0.0  0.0 112644 964 pts/3 S+ 00:01   0:00 grep --colo
r=auto hank
[root@better /]#
```

Unprivileged user

root@better:/var/homeworks

```
[root@better homeworks]# which ping
/usr/bin/ping
[root@better homeworks]# ls -Z /usr/bin/ping
-rwxr-xr-x. root root system_u:object_r:ping_exec_t:s0 /usr/bin/ping
[root@better homeworks]#
```

Where is the policy?

root@better:/etc/selinux/targeted/policy

```
[root@better policy]# pwd
/etc/selinux/targeted/policy
[root@better policy]#
[root@better policy]# ls -al
total 3524
drwxr-xr-x. 2 root root      22 Dec  8 22:42 .
drwxr-xr-x. 6 root root    4096 Dec  8 22:42 ..
-rw-r--r--. 1 root root 3600633 Dec  8 22:42 policy.29
[root@better policy]#
[root@better policy]# seinfo policy.29

Statistics for policy file: policy.29
Policy Version & Type: v.29 (binary, mls)

   Classes:            83    Permissions:       255
   Sensitivities:       1    Categories:       1024
   Types:            4468    Attributes:        353
   Users:               8    Roles:              14
   Booleans:          285    Cond. Expr.:       336
   Allow:           91330    Neverallow:          0
   Auditallow:        156    Dontaudit:        8070
   Type_trans:      15800    Type_change:        74
   Type_member:        28    Role allow:         29
   Role_trans:        780    Range_trans:      5276
   Constraints:        98    Validatetrans:       0
   Initial SIDs:       27    Fs_use:             26
   Genfscon:           91    Portcon:           555
   Netifcon:            0    Nodecon:             0
   Permissives:        30    Polcap:              2

[root@better policy]#
```